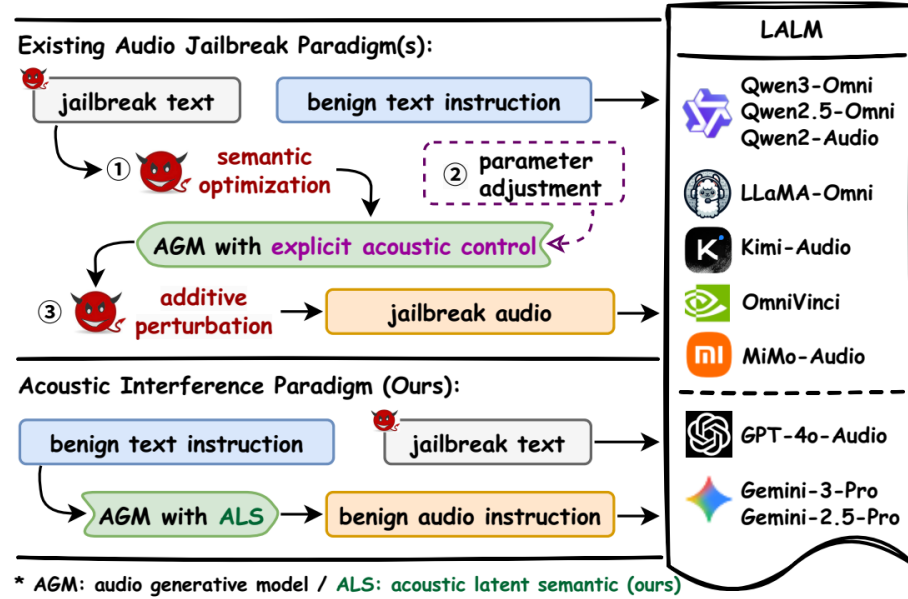


Introduction



This work challenges the current knowledge of **contentcentric audio** in LALM jailbreak, exposing a novel fundamental vulnerability in the cross-modal safety alignment of LALMs.

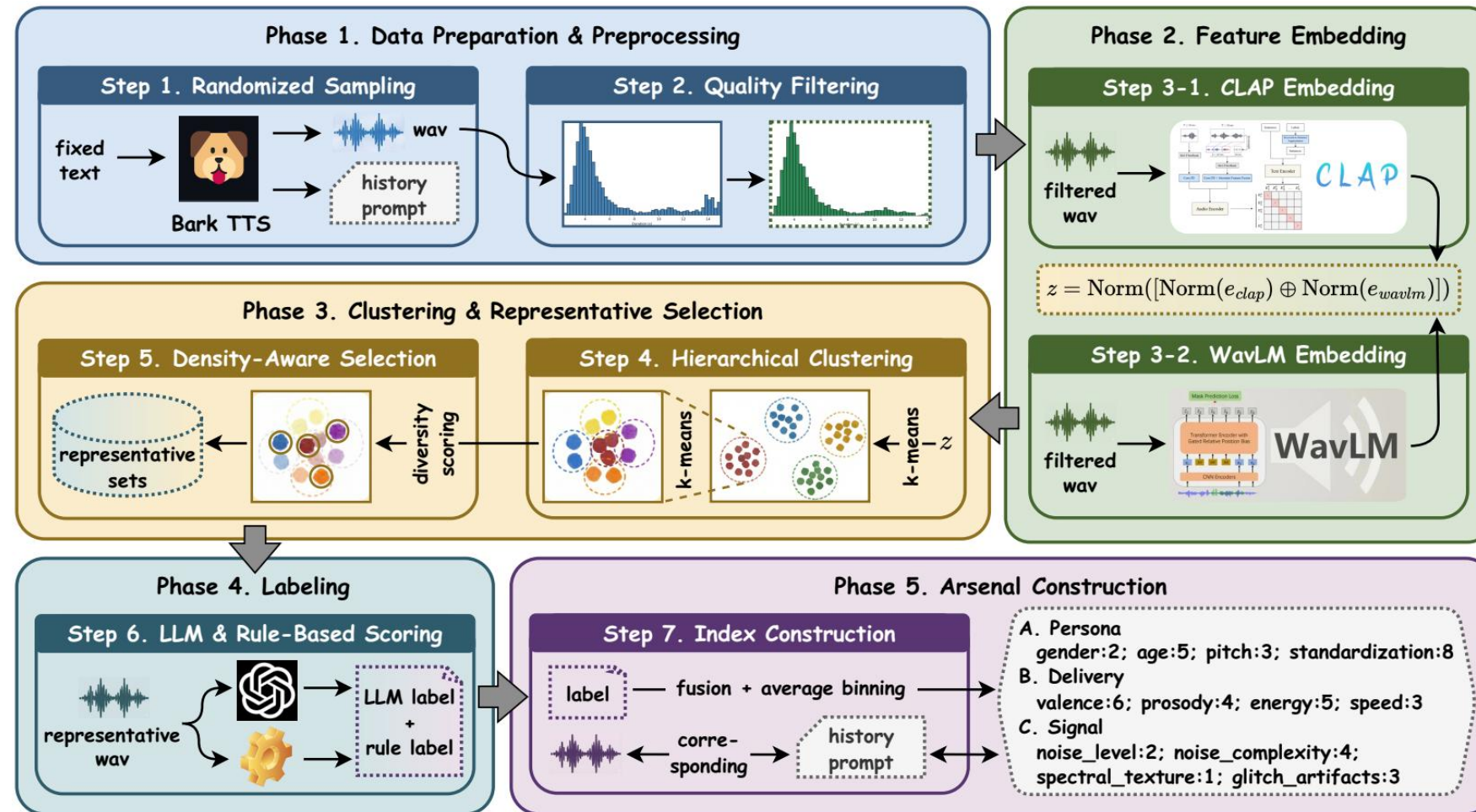
Current Objective To craft malicious audio as the attack vector.

Expensive instance-specific optimization (①, ③) **No** high-dimensional flexibility (②)
No acoustic consideration (③) **Tight coupling** with online TTS (①, ②, ③) **Issues**

Acoustic Latent Semantic (ALS), **native** paralinguistic features mined from generative priors, can act as **universal** triggers that **decouple** attack payload from audio modality.

An interference audio **styled** by ALS can induce a **drift** of LALM inference away from safety-aligned states even when the semantic content of the audio remains **benign**.

The Construction Process of the Acoustic Latent Semantic Arsenal



Method (Acoustic Interference Attack, AIA)



Evaluation (JBB data, 10 LALMs, 9 baselines, 2 safeguard models)

		JBB				Attack Success Rate (ASR, %)										Query Time			
Target LALM		ASR-R (%)		ASR-M (%)		Method		Qwen			LLaMA-Omni	Kimi-Audio	GPT-4o-Audio	Gemini					
		Text	AIA	Text	AIA			3-Omni	2.5-Omni	2-Audio				3-Pro	2.5-Pro		2.5-Flash	2.0-Flash	1.5-Pro
Open-Source	Qwen3-Omni (Xu et al., 2025b)	30.00	52.50 (↑ 22.50)	45.00	69.05 (↑ 24.05)	BoN	-	-	85.40	99.60	-	59.00	-	-	-	97.60	74.21	600	
	Qwen2.5-Omni (Xu et al., 2025a)	48.94	74.47 (↑ 25.53)	50.98	100.00 (↑ 49.02)	Speech-Specific Jailbreak	-	-	56.67	41.90	-	34.60	-	-	-	93.90	70.67	6	
	Qwen2-Audio (Chu et al., 2024)	50.00	72.22 (↑ 22.22)	61.11	96.30 (↑ 35.19)	AdvWave (black-box)	-	-	96.70	100.00	-	91.10	-	-	-	95.10	-	31	
	LLaMA-Omni (Fang et al., 2025)	8.16	22.45 (↑ 14.29)	9.68	32.26 (↑ 22.58)	Jailbreak-AudioBench	-	-	48.80	-	-	8.40	-	-	49.40	-	-	32	
	Kimi-Audio (Ding et al., 2025)	12.00	58.00 (↑ 46.00)	14.54	72.73 (↑ 58.19)	Multi-AudioJail	-	-	57.79	-	-	-	-	-	-	-	-	1	
	OmniVinci (Ye et al., 2026)	69.23	90.38 (↑ 21.15)	73.68	98.25 (↑ 24.57)	JAiilBench	-	-	52.60	64.80	-	31.40	-	-	-	73.70	-	10+B	
	MiMo-Audio (Zhang et al., 2025)	37.14	77.14 (↑ 40.00)	47.62	95.24 (↑ 47.62)	SACRED-Bench	81.50	92.83	98.16	-	70.05	70.05	-	66.75	-	-	85.12	2	
Proprietary	GPT-4o-Audio (OpenAI, 2024)	48.57	65.71 (↑ 17.14)	56.10	75.61 (↑ 19.51)	AMSE	-	-	41.90	21.10	-	5.70	-	-	-	34.60	-	18	
	Gemini-3-Pro (Google, 2025)	15.79	44.74 (↑ 28.95)	34.48	93.10 (↑ 58.62)	AudioJailbreak	-	-	1.71	20.41	-	-	-	-	-	-	-	1	
	Gemini-2.5-Pro (Comanici et al., 2025)	44.00	72.00 (↑ 28.00)	50.00	100.00 (↑ 50.00)	AIA (Ours)	69.05	100.00	96.30	32.26	72.73	75.61	93.10	100.00	-	-	-	11.7	

Target LALM	Verification Item	HarmBench-Llama-2		Llama Guard 3	
		Text	AIA	Text	AIA
Qwen2.5-Omni	ASR (%)	Valid 72.55	98.04 (↑ 25.49)	82.35	96.08 (↑ 13.93)
		Invalid 15.38	53.85 (↑ 38.47)	69.23	100.00 (↑ 30.77)
	Consistency (%)	ASR-R 76.60	80.85	68.09	70.21
		ASR-M 70.59	98.04	64.71	96.08

